

cf. Example 5 in Page 1050

$$\boxed{1} \quad \iiint_T y \, dV = \int_0^1 \int_{x/2}^{1-x/2} \int_0^{2-x-2y} y \, dz \, dy \, dx$$

$$= \int_0^1 \int_{x/2}^{1-x/2} y(2-x-2y) \, dy \, dx.$$

$$= \int_0^1 (2-x) \frac{1}{2} \left(\underbrace{(1-\frac{x}{2})^2 - (\frac{x}{2})^2}_{1-x} \right) - \frac{2}{3} \left((1-\frac{x}{2})^3 - (\frac{x}{2})^3 \right) dx$$

$$= \int_0^1 \left(1 - \frac{3}{2}x + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{2}{3} \left(-\frac{x^3}{4} + \frac{3}{4}x^2 - \frac{3}{2}x + 1 \right) \right) dx$$

$$= \int_0^1 \left(\frac{1}{6}x^3 - \frac{1}{2}x + \frac{1}{3} \right) dx = \frac{1}{24} - \frac{1}{4} + \frac{1}{3} = \frac{1}{8} "$$

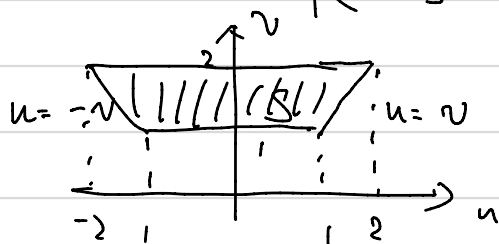
$\boxed{2.}$

$$\iint_R (x-y)^2 e^{-(x+y)(x-y)} \, dA$$



Set $\begin{cases} u = x+y \\ v = x-y \end{cases}$. Then $(x, y) = T(u, v) = \left(\frac{1}{2}(u+v), \frac{1}{2}(u-v) \right)$

and R is the image of S w.r.t. T , where



(See Example 3 in page 114)

$$\therefore \iint_R (x-y)^2 e^{-(x+y)(x-y)} \, dA = \iint_S v^2 e^{-uv} \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$$

$$\left(\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2} \right)$$

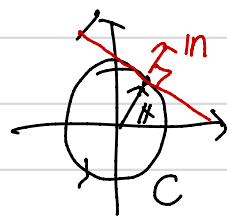
$$= \frac{1}{2} \int_{-2}^2 \int_{-v}^v v^2 e^{-uv} \, du \, dv = \frac{1}{2} \int_{-2}^2 \left[-v e^{-uv} + v e^{uv} \right] dv$$

$$= \frac{1}{4} \left[e^{-v^2} + e^{v^2} \right]_{-2}^2 = \frac{1}{4} (e^{-4} + e^4 - e^{-1} - e)$$

[3]. $r(t) = \langle 2\cos t, 2\sin t \rangle, \quad 0 \leq t \leq 2\pi$

$$\begin{aligned}
 (i) \quad \int_C \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} \vec{F}(r(t)) \cdot r'(t) dt \\
 &= \int_0^{2\pi} \frac{\langle 2\sin t, -2\cos t \rangle \cdot \langle -2\sin t, 2\cos t \rangle}{4} dt \\
 &= \int_0^{2\pi} (-1) dt = -2\pi
 \end{aligned}$$

(ii) Note that $\vec{n} = \frac{x\vec{i} + y\vec{j}}{\sqrt{x^2 + y^2}}$



Then if $(x, y) \in C$,

$$\vec{F} \cdot \vec{n} = \frac{(y\vec{i} - x\vec{j}) \cdot (x\vec{i} + y\vec{j})}{(x^2 + y^2)^{3/2}} = 0$$

$$\therefore \int_C \vec{F} \cdot \vec{n} dV = \int_C 0 dV = 0.$$

[4] Note that $\text{Curl } \vec{F} = \vec{0}$ in $\mathbb{R}^3$, hence $\vec{F}$ is conservative. Let's find a potential function f so that $\vec{F} = \nabla f$. Then

$$\int_C \vec{F} \cdot d\vec{r} = f(r(\pi)) - f(r(0)),$$

$$\begin{aligned}
 \text{where } r(t) &= \langle \sin t, t, 2t \rangle \\
 &= f(0, \pi, 2\pi) - f(0, 0, 0)
 \end{aligned}$$

• Since $\frac{\partial f}{\partial x} = \sin y$, $f(x, y, z) = x \sin y + g(y, z)$

• Since $\frac{\partial f}{\partial y} = x \cos y + \frac{\partial g}{\partial y} = x \cos y + \cos z$, for some $g(y, z)$

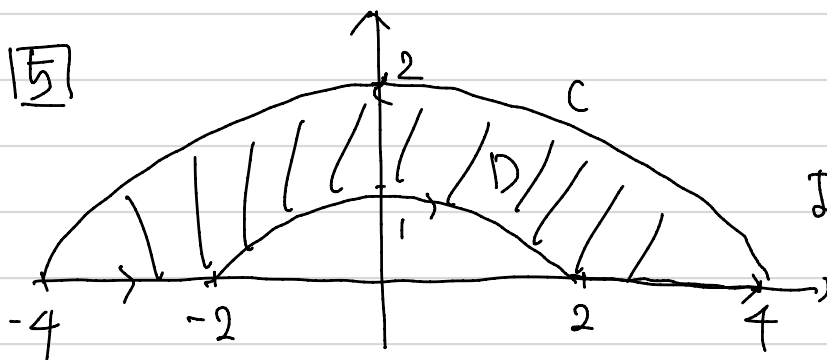
$g(y, z) = y \cos z + h(z)$ for some $h(z)$ and

$f(x, y, z) = x \cos y + y \cos z + h(z)$

• Since $\frac{\partial f}{\partial z} = -y \sin z + h'(z) = 1 - y \sin z$

$h(z) = z + C$ and $f(x, y, z) = x \cos y + y \cos z + z + C$ for some C .

$\therefore \int_C F \cdot dr = f(0, \pi, 2\pi) - f(0, 0, 0) = 3\pi$



By Green's theorem,

$$I = \oint_C y^2 dx + 3xy dy$$

$$= \iint_D (3y - 2y) dA$$

$$= \iint_D y dA$$

Let $\begin{cases} x = 2r \cos \theta \\ y = r \sin \theta \end{cases}$, $1 \leq r \leq 2$, $0 \leq \theta \leq \pi$

Then D is the image of $S = \{(r, \theta) : 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$ w.r.t. $T(r, \theta) = (2r \cos \theta, r \sin \theta)$

$$\therefore I = \iint_S r \sin \theta \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dA,$$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} 2 \cos \theta & \sin \theta \\ -2r \sin \theta & r \cos \theta \end{vmatrix} = 2r$$

$$= \int_1^2 \int_0^\pi r \sin \theta \cdot 2r \, dr \, d\theta$$

$$= \int_1^2 2r^2 \, dr \int_0^\pi \sin \theta \, d\theta = \frac{2}{3} (8-1) \cdot 2 = \frac{28}{3}.$$

16 See page 1168.

$$\underline{17} \quad A = \iint_R \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dA,$$

where $z = \sqrt{2xy}$ and $R = \{(x, y) : 1 \leq x \leq 4, 1 \leq y \leq 4\}$

$$= \iint_R \sqrt{\left(\frac{\sqrt{2y}}{2\sqrt{x}}\right)^2 + \left(\frac{\sqrt{2x}}{2\sqrt{y}}\right)^2 + 1} \, dA$$

$$= \iint_R \sqrt{\frac{(x+y)^2}{2xy}} \, dA = dA = \frac{1}{\sqrt{2}} \iint_R \left(\frac{\sqrt{y}}{\sqrt{x}} + \frac{\sqrt{x}}{\sqrt{y}}\right) dA$$

by symmetry

$$= \frac{2}{\sqrt{2}} \int_1^4 \int_1^4 \frac{\sqrt{y}}{\sqrt{x}} \, dy \, dx = \sqrt{2} \left(\int_1^4 \sqrt{y} \, dy \right) \left(\int_1^4 \frac{1}{\sqrt{x}} \, dx \right)$$

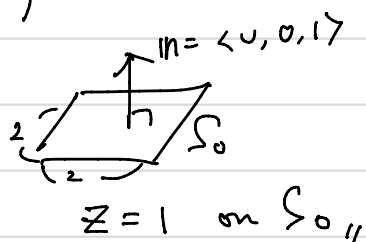
$$= \sqrt{2} \cdot \frac{2}{3} (8-1) \cdot 2 (2-1) = \frac{28}{3} \sqrt{2}.$$

10] By the divergence theorem,

$$\iint_{S \cup S_0} \mathbf{F} \cdot d\mathbf{S} = \iiint_R \operatorname{div} \mathbf{F} dV = - \iiint_R 1 = -8$$

$$\partial R = \operatorname{div} \mathbf{F} = (1 + 2x) + (1 - 2x) - 3 = -1$$

$$\iint_{S_0} \mathbf{F} \cdot d\mathbf{S}$$



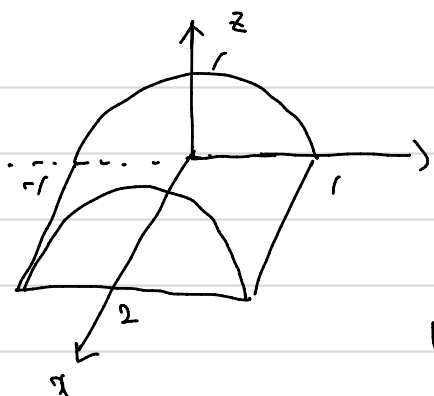
$$= \iint_{S_0} \langle x + x^2, y - 2xy, -3 \rangle \cdot \langle 0, 0, 1 \rangle dS$$

$$= \iint_{S_0} -3 dS = -3 \operatorname{Area}(S_0) = -12$$

$$\therefore \iiint_R \mathbf{F} \cdot d\mathbf{S} = - \iint_{S_0} \mathbf{F} \cdot d\mathbf{S} - 8$$

$$= 4$$

19



By divergence theorem,

$$I = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_D \operatorname{div} \mathbf{F} \, dV$$

$$\text{where } D = \{ (x, y, z) : 0 \leq x \leq 2, -1 \leq y \leq 1, 0 \leq z \leq \sqrt{1-y^2} \}$$

$$\operatorname{div} \mathbf{F} = 4x + 2y + 2z$$

$$\therefore I = \int_0^2 \int_{-1}^1 \int_0^{\sqrt{1-y^2}} (4x + 2y + 2z) \, dz \, dy \, dx$$

$$= \left(\int_0^2 4x \, dx \right) \left(\int_{-1}^1 \int_0^{\sqrt{1-y^2}} 1 \, dz \, dy \right)$$

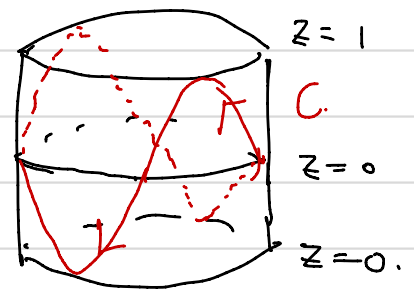
$$+ 4 \int_{-1}^1 \underbrace{y \sqrt{1-y^2} \, dy}_{\text{odd function}} + 4 \int_{-1}^1 \int_0^{\sqrt{1-y^2}} z \, dz \, dy$$

$$= 8 \cdot \frac{\pi}{2} + 0 + 2 \int_{-1}^1 \underbrace{(1-y^2)}_{\text{even}} \, dy$$

$$= 4\pi + 4 \left[y - \frac{1}{3}y^3 \right]_0^1 = \frac{8}{3} + 4\pi //$$

10 Note that C is the closed curve on the surface $z = 2xy$. By Stokes' theorem

$$I = \iint_S \text{curl } \vec{F} \cdot d\vec{S}$$



where S is surface $z = 2xy$, $(x, y) \in D$

with $D = \{(x, y) : x^2 + y^2 \leq 1\}$,

$$\text{and } \text{curl } \vec{F} = \langle -2z, -4x^2, -1 \rangle$$

Here, the orientation of surface S should be taken as the downward unit normal vector.

$$\therefore I = \iint_D \langle -4xy, -4x^2, -1 \rangle \cdot \langle 2y, 2x, -1 \rangle dA$$

$$= \iint_D (-8xy^2 - 8x^3 + 1) dA$$

Polar coordinate

$$= \int_0^{2\pi} \int_0^1 (-8r^3 \cos \theta \sin^2 \theta - 8r^3 \cos^3 \theta + 1) r dr d\theta$$

$$= \int_0^{2\pi} \int_0^1 (-8r^4 \cos \theta + r) dr d\theta$$

$$= 2\pi \int_0^1 r dr = \pi //$$